

Silver Claims ODS — Data Model

A canonical, multi-source claims integration model — unified across ACO FHIR and Non-ACO proprietary sources. Designed for structural fidelity, pipeline idempotency, and clean Medallion layer boundaries.

SILVER ODS LAYER



AGENDA

What We'll Cover

1. Problem Statement Two source worlds, one model needed

2. Architecture Overview Bronze → Silver Raw → Silver ODS → Gold

3. Silver Raw Layer What it does and does not do

4. Silver Raw ER Diagram Source-aligned entity relationships

5. Core Transformations Keys, mapping, CDC, normalization

6. Silver ODS ER Diagram Member-centric entity relationships

7. Gold Layer Reporting & analytics (placeholder)

8. Summary Key takeaways (placeholder)

Problem Statement: Two Source Worlds, One Model Needed

Claims data arrives from two structurally different source systems. Without a unified integration layer, downstream analytics are fragmented, inconsistent, and unreliable.

Non-ACO — Proprietary Format

Claim type drives separate header/line table pairs with no unified grain:

- IP Header / IP Line
- OP Header / OP Line
- Pharmacy Header / Pharmacy Line
- Dental Header / Dental Line
- Vision Header / Vision Line

ACO — FHIR Source

- FHIR v4 ExplanationOfBenefit (EOB) resources via BCDA
- Nested JSON structure — requires flattening at ingest
- Bundles can be inconsistent; `source_claim_id` is the reliable anchor

FHIR IDs alone are not reliable. `source_claim_id` is always the anchor.

Medallion Architecture: Four Layers, One Purpose

Each layer has a distinct, non-overlapping responsibility. Business logic does not leak downward. Raw noise does not leak upward.

Bronze

Raw landing zone. Exact copy of source files. No transformation. Append-only. No dedup. No schema enforcement.

Silver Raw

Standardized ingestion layer. Technical dedup via ROW_NUMBER(). record_hash for CDC. Source-aligned keys. ACO + Non-ACO preserved as-is. No business logic.

Silver ODS

Canonical integration layer. Surrogate keys. Member-centric. Structural transformations only: patient→member mapping, adjudication scope, bridge normalization. Source traceability preserved.

Gold

Reporting and analytics layer. Conformed dimensions. Flattened models. CCLF export mapping. NOT part of this model.

BRONZE → SILVER RAW

From Bronze to Silver Raw — Unifying Two Worlds

Two structurally different sources. One normalized, source-aligned output.

ACO Source

FHIR

- FHIR
ExplanationOfBenefit
- Nested JSON
- item[], adjudication[],
diagnosis[]
- source_claim_id anchor

Non-ACO Source

FLAT FILE

- Proprietary flat files
- Wide table structure
- diagnosis_1..n,
procedure_1..n
- paid_amt, copay_amt,
deductible_amt

Ingest & Normalize

1. EXPLODE — flatten
FHIR arrays into rows
2. UNPIVOT — normalize
repeated columns into
rows
3. STANDARDIZE — unify
keys: claim_id,
patient_id, line_number

Output Tables

- claim_header
- claim_line
- claim_adjudication
- claim_diagnosis
- claim_procedure
- claim_provider

Column-Level Source Mapping

Concept	ACO Source	Non-ACO Source	Silver Raw
Claim ID	eob.id / identifier[].value	claim_id (header table)	source_claim_id
Patient ID	eob.patient.reference	member_id / patient_id	patient_id
Claim Line	eob.item[].sequence	claim_line_number	line_number
Diagnosis	eob.diagnosis[].diagnosisC odeableConcept	diagnosis_1..n	claim_diagnosis (rows)
Procedure	eob.procedure[].procedure CodeableConcept	procedure_1..n	claim_procedure (rows)
Adjudication	eob.item[].adjudication[]	paid_amt, copay_amt, deductible_amt	claim_adjudication (rows)

Silver Raw: Standardize Without Transforming

Silver Raw is the first layer where data gets a shape. It does not interpret data — it preserves it safely.

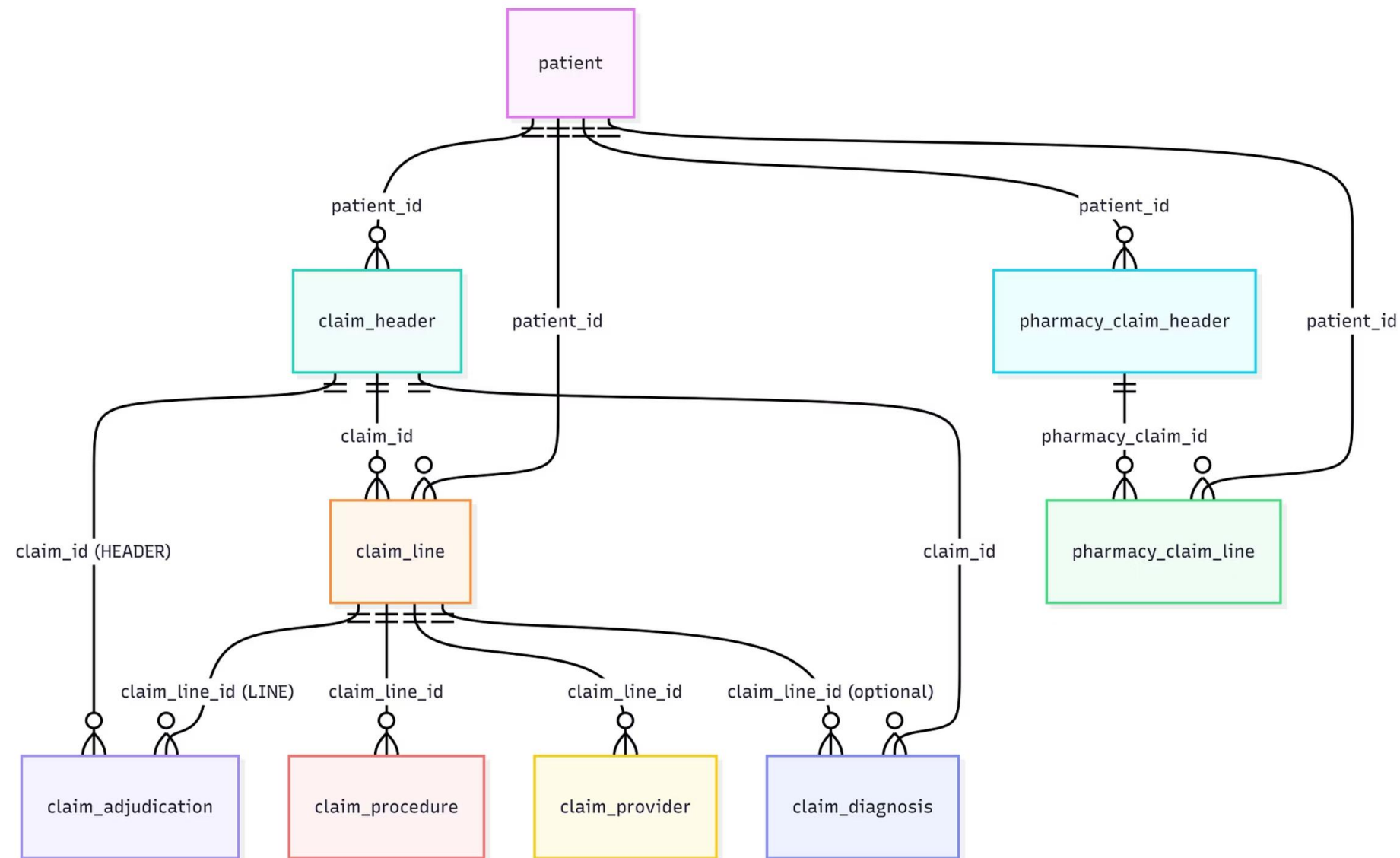
✓ What Silver Raw DOES

- Enforces schema per source type (CSV, FHIR JSON)
- Technical dedup via ROW_NUMBER() on source_claim_id
- Generates record_hash for downstream CDC
- Stamps ingestion_timestamp, batch_id, source_file_id on every row
- Carries patient_id as-is from source
- Preserves all source field names and raw values

✗ What Silver Raw Does NOT Do

- No patient → member resolution
- No surrogate key generation
- No cross-source joins
- No business deduplication
- No code standardization
- No analytical logic of any kind

Silver Raw: Entity Relationships



Core Transformations: Silver Raw to Silver ODS

Six structural transformations at this boundary. None are analytical — all are integration-layer responsibilities.

Surrogate Key Generation

Generated PKs on every fact table.
source_claim_id is always retained
alongside.

Patient → Member Mapping

patient_id resolves to member_wid
via patient_member_map.
dim_member is the central entity.

Business Dedup + CDC

Full CDC/Merge replaces Raw's
ROW_NUMBER() dedup.
record_hash drives change
detection.

Bridge Table Normalization

Flat repeated columns become
bridge tables: diagnosis, procedure,
provider — each with a FK back to
the parent claim.

Adjudication Scope Enforcement

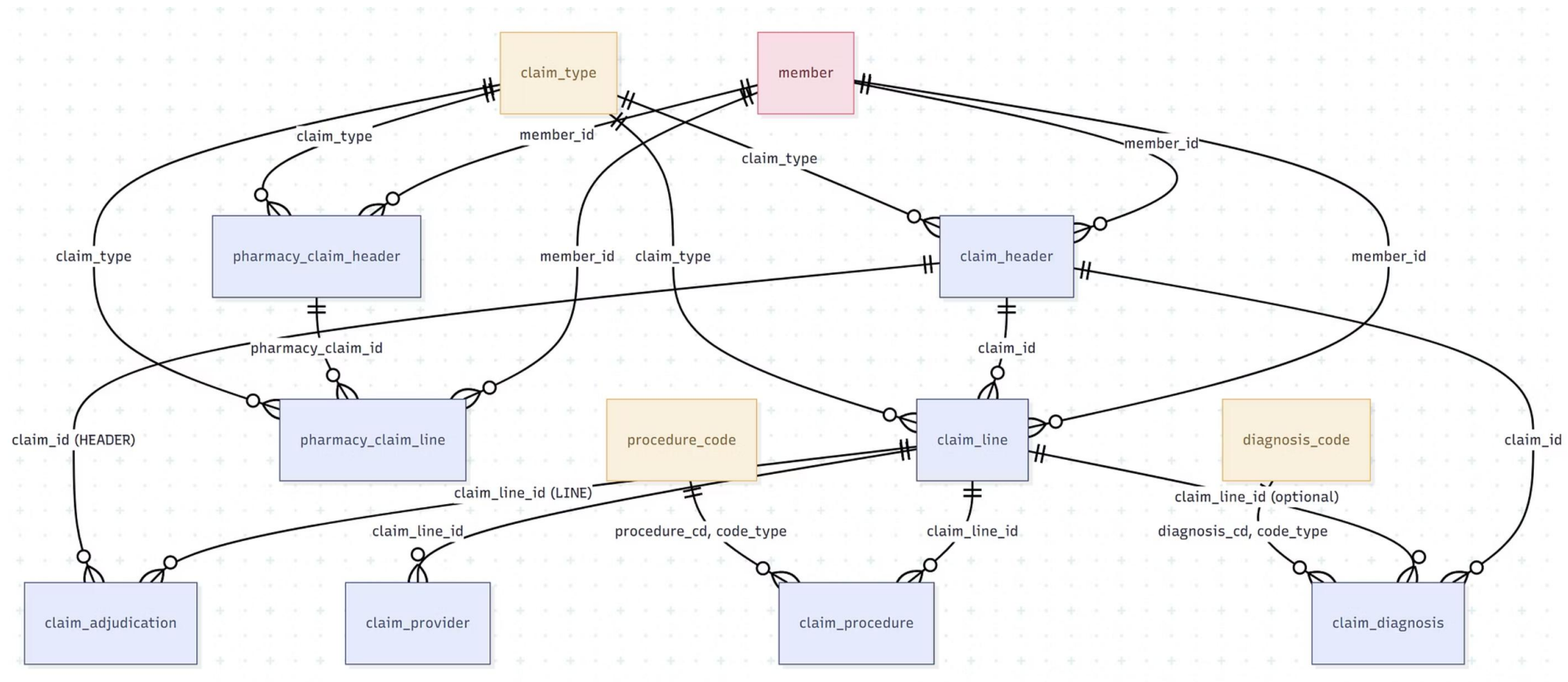
adjudication_scope = LINE or
HEADER. Exactly one FK populated
per row — never both.

Code Standardization

Raw codes preserved as-is.
Standardized columns are added
alongside; Silver ODS never
overwrites source values.

Silver ODS: Entity Relationships

Member-centric, surrogate-keyed relational model. Bridge tables for diagnosis, procedure, and provider. All relationships are logical — no DB REFERENCES enforced.



Gold Layer — Reporting & Analytics

[Placeholder — Gold layer content to be added]

What Goes Here

- Conformed dimensions (dim_member, dim_provider, dim_date)
- Flattened, denormalized models for reporting
- CCLF export mapping
- Aggregated financial summaries (allowed, paid, billed)
- ACO performance metrics

Key Principles

- Built on top of Silver ODS — never directly on Bronze or Silver Raw
- Analytical logic lives here, not in Silver
- Conformed keys replace surrogate keys
- Optimized for query performance

📄 ⚠️ This slide is a placeholder. Gold layer design is out of scope for this model — content to be completed separately.

SUMMARY

Summary & Key Takeaways

[Placeholder — Summary content to be added]

[Takeaway 1] Placeholder — e.g. Silver ODS is the canonical integration layer, not a reporting layer.

[Takeaway 2] Placeholder — e.g. source_system + source_claim_id is the universal natural key across all sources.

[Takeaway 3] Placeholder — e.g. Adjudication table is the source of truth for financial adjustments.

[Takeaway 4] Placeholder — e.g. All analytical logic belongs in Gold — never in Silver.

 **⚠** This slide is a placeholder. Add final takeaways and next steps before presenting.